

ENGINE EVIDENCE

Four figures. The scope of each.

Internal tests on synthetic fixtures. Each figure has its own workload and measurement boundary. These are not full product performance guarantees or an external audit.

500K entities under 2 seconds

500K entities under 2 seconds on the fixed synthetic fixture, with about 55K transactions. This is a detection test, not the full product cycle.

alert lifecycle under 10 ms

Alert lifecycle under 10 ms per alert, from a detection result to a persisted, queryable alert. Batch results are amortised per alert.

federation bilateral round under 10 seconds

Federation bilateral round under 10 seconds on TCP loopback development transport. Semi-honest security. This does not measure a real inter-bank network.

VRAM under 1 GB

VRAM under 1 GB in the measured detection-pass working set after warm-up. Resident ingestion and model inference are excluded. This is not total-system memory.

Read the boundaries together

The timing fixture has about 55K transactions. The memory fixture is a separate detection-pass measurement. Neither proves full-system performance at a realistic transaction density with all components resident. The memory regression gate is 1 GiB; the recorded detection-pass working set is below 1 GB.

Methods and limitations: zquas.ai/benchmark.html

LOCAL MONITORING

One synthetic walkthrough.

All people, accounts and payments here are fictional. The monitor has not yet been measured on the full synthetic worlds. No detection rates or alert counts are claimed.

Several payers in | Synthetic example

Several synthetic payers send money into one account. The bank uses the incoming payments in its own journal.

Money out within a day | Synthetic example

Money is debited out within a day of arriving. An outgoing debit is not proof that another bank received it.

The pattern score | Synthetic example

The local monitor scores a funnel or rapid pass-through pattern using its own observations and frozen calibration. With sufficient history, it combines movement with context from earlier activity. The score can nominate work for review. It does not establish that a crime occurred.

The analyst queue | Synthetic example

Nominations are ranked. Admission depends on the daily analyst-time limit and monthly investigation allowance. This local monitoring step does not hold a payment.

The signed tm-alert line | Synthetic example

The monitor writes a tm-alert line for a new queue entry, admission or expiry. Review and refusal decisions carry a proof, and the institution signs the log head. The supervisor can check the record using the public key. Synthetic data, development keys.

The bank decides | Synthetic example

The analyst examines the bank's own evidence and follows its procedures. The bank decides what action is justified. The pattern alone does not establish wrongdoing.

Read the walkthrough: zquas.ai/use-case.html

What is established. What remains to measure.

Internal evidence

The engine figures were checked against the F1_ENGINE code, claim register and retained reports. The fixed 500K detection fixture, per-alert handling, loopback federation and detection-pass memory each have a different scope. They must not be combined into a full-system performance promise.

Signed records

The published supervisor check verified the synthetic logs in run R43.0 B, using development keys. The local monitor also records its queue decisions as tm-alert lines under signed log heads. Verification establishes the integrity of that record, not the correctness of a criminal allegation.

Local monitor

The walkthrough follows kit tm-run, its frozen feature sheet and the current queue and logging implementation. Full-world detection rates and alert counts are not yet measured. This briefing does not publish them.

Network phase

Current federation security is semi-honest: participants are assumed to follow the protocol. Deliberate protocol deviation is outside that security claim. The optional network phase needs separately agreed sharing rules, legal review and operational measures.

Exchange participation

Exchange participation is being retested with information available to each exchange at the decision time.

Further reading

zquas.ai/benchmark.html
zquas.ai/use-case.html
zquas.ai/founding-partner.html
zquas.ai/contact.html

ZQUAS B.V., the Netherlands. This is a pilot discussion document. Terms and scope are subject to agreement.